Fundamentals Chapter 2

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1a. Assume m and n are positive. Then g^{nm} is $g \cdot g \cdots g$ where g is multiplied nm times. Also, $(g^n)^m = (g \cdot g \cdots g) \cdots (g \cdot g \cdots g)$ where g is multiplied n times and then the result is multiplied m times. There are totally nm gs being multiplied, so by associativity, it is the same as $g \cdot g \cdots g = g^{nm}$.

If $m = -1$, then $g^{nm} = g^{-n}$. Since $g \cdot g^{-1} = 1$, $g^n \cdot (g^{-1})^n = 1$. Since $g^n \cdot (g^n)^{-1} = 1$, we have that $(g^{-1})^n = g^{-n} = (g^n)^{-1}$. Similarly, if $n = -1$, $g^{nm} = g^{-m} = (g^{-1})^m$.

If m is negative, then $g^{nm} = g^{-n(-m)} = (g^{n(-m)})^{-1} = ((g^n)^{-m})^{-1} = (g^n)^m$.

1b. Assume H is closed under multiplication and inverses. Then for all $g, h \in H$, $h^{-1} \in H$ and $gh^{-1} \in H$.

Now, assume $gh^{-1} \in H$ for all $g, h \in H$. Since h^{-1} is defined for all $h \in H$, it is closed under inverses. Since multiplication of g by h^{-1} is defined for all $g, h \in H$, it is closed under multiplication.

3. Let $\cdot$ be the operation for H and $\star$ be for G. Since f is a homomorphism, $f(h_1 \cdot h_2) = f(h_1) \star f(h_2)$ for all $h_1, h_2 \in H$. Let $h \in H$. Then $f(h \cdot e_H) = f(h) \star f(e_H)$. This gives us $f(h) = f(h) \star f(e_H)$, so $f(e_H) = e_G$.

Also, $f(h \cdot h^{-1}) = f(h) \star f(h^{-1})$, so $f(e_H) = e_G = f(h) \star f(h^{-1})$. This means $f(h)^{-1} = f(h^{-1})$.

4. Let the operation for G be $\cdot$ and G' be $\star$. Assume there exists a homomorphism $f : G \to G'$. Let $g \in G$ map to $g' \in G'$. Then $f(g \cdot g) = g' \star g'$, or $f(g^2) = g'^2$. Similarly, $f(g^n) = g'^n$

Let the order of g be n. Then $g \cdot g^{n-1} = e_G$. But, if $g' \neq e_{G'}$, then $g' \star g'^{n-1} = g'^n$, and $f(e_G) \neq g'^n$, which is a contradiction. So the only possibility is where $g' = e_{G'}$, so every element in G is mapped to $e_{G'}$.

5. Let $g_1 = (x_1, y_1)$ and $g_2 = (x_2, y_2)$ so that $g_1, g_2 \in G$. Define $\cdot$ as $g_1 \cdot g_2 = (x_1 + x_2, y_1 + y_2)$. Also, let $h_1, h_2 \in H$. Define $\star$ as $h_1 \star h_2 = h_1 + h_2$.

Then $(G, \cdot) \cong (H, \star)$. Since $|\mathbb{R}^2 \backslash \{(0,0)\}| = |\mathbb{C} \backslash \{0\}|$, all that remains to show is that there exists f such that $f(g_1 \cdot g_2) = f(g_1) \star f(g_2)$ for all $g_1, g_2 \in G$.

Define $f : G \to H$ so that $f((x, y)) = x + iy$. Then

$$f(g_1 \cdot g_2) = f((x_1 + x_2, y_1 + y_2)) = (x_1 + x_2) + i(y_1 + y_2)$$

$$= (x_1 + iy_1) + (x_2 + iy_2) = f(g_1) + f(g_2),$$

as desired.

7. The inverse of ab is $b^{-1}a^{-1}$ because $abb^{-1}a^{-1} = aa^{-1} = e$.

The inverse of $a_1 \cdots a_n$ is $a_n^{-1} \cdots a_1^{-1}$ because

$$a_1 \cdots a_n a_n^{-1} \cdots a_1^{-1} = a_1 \cdots a_{n-1} a_{n-1}^{-1} \cdots a_1^{-1} = \cdots = e.$$

8. Reflexivity: let $(G, \cdot)$ be a group. To create a homomorphism from $G \to G$, let f map each element in G to itself. Then, for $g_1, g_2 \in G$, $f(g_1 \cdot g_2) = g_1 \cdot g_2 = f(g_1) \cdot f(g_2)$. This along with the fact that $|G| = |G|$ means that $G \cong G$ and $\cong$ is reflexive.

Symmetry: let $(H, \cdot), (G, \star)$ be groups. Assume there is an isomorphism from $H \to G$ and let the homomorphism be f. Since f is a bijection, it has an inverse f^{-1}. Let $\cdot$ be the binary operation for H and $\star$ be for G. For all $h_1, h_2 \in H$, we know that

$$f(h_1 \cdot h_2) = f(h_1) \star f(h_2),$$

$$f^{-1}(f(h_1 \cdot h_2)) = f^{-1}(f(h_1) \star f(h_2)),$$

$$h_1 \cdot h_2 = f^{-1}(f(h_1) \star f(h_2)).$$

Let $g_1 = f(h_1)$ and $g_2 = f(h_2)$. Simplifying the equation gives us

$$f^{-1}(g_1) \cdot f^{-1}(g_2) = f^{-1}(g_1 \star g_2).$$

So f^{-1} is a homomorphism $G \to H$. Since $|G| = |H|$, $G \cong H$. So $H \cong G$ implies $G \cong H$ and $\cong$ is symmetric.

Transitivity: let $(G, \cdot), (H, \times), (K, \star)$ be groups. Assume $G \cong H$ and $H \cong K$ and let the homomorphism from $G \to H$ be f and from $H \to K$ be g. Since f, g are bijections, define $h : K \to G$ so that $h = (f \circ g)^{-1}$. We know the following properties to be true for all $g_1, g_2 \in G$ and $h_1, h_2 \in H$:

$$f(g_1 \cdot g_2) = f(g_1) \times f(g_2),$$

$$g(h_1 \times h_2) = g(h_1) \star g(h_2)$$

Our goal is to show that

$$h(k_1 \star k_2) = h(k_1) \cdot h(k_2)$$

for all $k_1, k_2 \in K$. Let $h_1 = f(g_1)$, $h_2 = f(g_2)$, $k_1 = g(h_1)$, $k_2 = g(h_2)$. Then $g_1 = h(k_1)$ and $g_2 = h(k_2)$. Substituting these into an earlier equation gives us

$$g(f(g_1) \times f(g_2)) = g(f(g_1)) \star g(f(g_2)).$$

We can simplify the right-hand side to $k_1 \star k_2$. Taking h of both sides gives us

$$f^{-1}(g^{-1}(g(f(g_1) \times f(g_2)))) = h(k_1 \star k_2),$$

$$f^{-1}(f(g_1) \times f(g_2)) = h(k_1 \star k_2).$$

From earlier, we know that $f(g_1) \times f(g_2) = f(g_1 \cdot g_2)$, so the left-hand side becomes

$$f^{-1}(f(g_1 \cdot g_2)) = g_1 \cdot g_2 = h(k_1) \cdot h(k_2).$$

Therefore, we have

$$h(k_1) \cdot h(k_2) = h(k_1 \star k_2),$$

as desired.

9. Since there exists $C \in \mathbb{N}$ such that if $n < -C$ or $n > C$ then $f(n) = 0$, $\sum_{n=-\infty}^{\infty} f(n)g(x-n) = \sum_{n=-C}^{C} f(n)g(x-n)$ so it is a finite sum.

To show that $(S, *)$ satisfies all the properties of a group except the existence of inverses, we must show closure of $*$, the existence of an identity element, and associativity.

To show closure of $*$, we must show that if $f, g \in S$ then $f * g \in S$. We know that $f(x) = 0$ when $x > |C_f|$ and $g(x) = 0$ when $x > |C_g|$ for some $C_f, C_G \in \mathbb{N}$. So for $(f * g)(x)$ to be nonzero, both $f(n)$ and $g(x-n)$ must be nonzero for some n. So

$$|n| \leq C_f, \quad |x - n| \leq C_g.$$

Combining these inequalities gives us

$$|x| < |n| + |x - n| \leq C_f + C_g.$$

Therefore, $f(n)g(x-n) = 0$ for all n when $|x| > C_f + C_g$, so $(f * g)(x) = 0$ when $|x| > C_f + C_g$. This means $f * g \in S$.

Define $e : \mathbb{Z} \to \mathbb{R}$ by $e(x) = 1$ if $x = 0$ and $e(x) = 0$ otherwise. The function e clearly belongs to S since it is non-zero only at $k = 0$. We know that

$$(f * e)(x) = \sum_{n=-\infty}^{\infty} f(n)e(x-n).$$

Since $e(x-n) = 0$ unless $x - n = 0$ (or $x = n$), the sum simplifies to

$$(f * e)(x) = f(x) \cdot e(0) = f(x)$$

so e acts as an identity element in S.

To show associativity, we need to show that $(f * g) * h = f * (g * h)$ for all $f, g, h \in S$. Consider the left-hand side:

$$((f*g)*h)(x) = \sum_{m=-\infty}^{\infty} (f*g)(m)h(x-m) = \sum_{m=-\infty}^{\infty} \left(\sum_{n=-\infty}^{\infty} f(n)g(m-n) \right) h(x-m).$$

Swapping the sums, we get

$$\sum_{n=-\infty}^{\infty} f(n) \left(\sum_{m=-\infty}^{\infty} g(m-n)h(x-m) \right).$$

Letting $k = m - n$,

$$= \sum_{n=-\infty}^{\infty} f(n) \left(\sum_{k=-\infty}^{\infty} g(k)h(x-n-k) \right).$$

Now, consider the right-hand side:

$$(f*(g*h))(x) = \sum_{n=-\infty}^{\infty} f(n)(g*h)(x-n) = \sum_{n=-\infty}^{\infty} f(n) \left(\sum_{k=-\infty}^{\infty} g(k)h(x-n-n) \right)$$

which is identical to the left-hand side.

10. We need to show that $f(a+b) = f(a) + f(b)$ implies that f maps x to nx for some constant n. Let $b = a$. Then we have $f(2a) = 2f(a)$. Now assume that $f(ka) = kf(a)$ for all $k \in \mathbb{Z}$. From the equation above, we have

$$f((k+1)a) = f(ka + a) = f(ka) + f(a) = (k+1)f(a),$$

so by induction, $f(ka) = kf(a)$ for all $k \in \mathbb{Z}$.

Letting $a = 1$, we have $f(k) = kf(1)$, and since $f(1)$ is constant, we can let $n = f(1)$. So for any x, $f(x) = nx$.

11. A 9×9 Rubik's cube can be treated as a 3×3 by rotating either the top three, middle three, or bottom three layers at once. So for any movement of a 3×3 Rubik's cube, we can do the same movement on three layers at a time in a 9×9 Rubik's cube.

13. Assume that f is injective. Then if $f(h_1) = f(h_2)$ then $h_1 = h_2$. So the only element that gets mapped to e_G is e_H, so $\ker(f) = \{e_H\}$.

Now, assume that $\ker(f) = \{e_H\}$. Then the only element in H that gets mapped to e_G is e_H. Assume that $f(a) = f(b)$ for some $a, b \in H$. We need to show that $a = b$.

Multiplying both sides by $f(b)^{-1}$ (on the left), we get $f(b)^{-1}f(a) = f(b^{-1}a) = e_G$. This implies that $b^{-1}a \in \ker(f)$, so $b^{-1}a = e_H$. Multiplying both sides by b on the left, we get $a = b$. Therefore, f is injective.